CHAPTER 1: INTRODUCTION AND OBJECTIVE

1.1 Introduction:

The **School Management System** is a C++ based application designed to manage and organize different school activities. It includes different users such as:

I. **Super Admin:** Manages schools and principals.
II. **Principal:** Manages teachers and school activities.
III. **Teacher:** Manages students, attendance and results.
IV. **Student:** Accesses student-related information.
V. **Parent:** Accesses information related to their child.

The system reduces manual work and helps maintain school records in an organized and efficient way.

1.2 Project Profile:

- Project Title: School Management System
- Platform: Windows
- Project Type: Application Software
- Category: Text-based interface / File Handling
- Language: C++

1.3 Objectives:

- To reduce manual work in managing school records.
- To store and organize information efficiently.
- To manage different users through separate roles.
- To make records easy to access and update.
- To reduce errors and improve accuracy.
- To apply C++ and file handling concepts practically.

1.4 Existing Manual System:

In the manual system, school information such as student details, attendance, results and staff records is maintained using registers and files. Searching and updating these records manually can be difficult and time-consuming.

1.4.1 Limitations of the Existing Manual System:

- Time-consuming and inefficient.
- Prone to human errors.
- Difficult to store, retrieve, and update data.
- Hard to analyze large amounts of information.
- Lack of consistency and standardization.

1.5 Proposed System:

- The proposed system computerizes school management activities.
- It provides separate access for different users.
- It stores and retrieves records using file handling.
- It reduces paperwork and manual effort.

1.5.1 Advantages of Proposed System:

- Saves time and increases efficiency.
- Reduces errors and improves accuracy.
- Organizes records for easy access.
- Reduces paperwork.
- Provides role-based access.
- Easy to use and manage.

CHAPTER 2: REQUIREMENT ANALYSIS

2.1 Introduction:

This chapter discusses the software, hardware and training requirements needed for developing and running the **School Management System**.

2.2 Software Requirements:

- ✓ **Operating System:** Windows.
- ✓ **Programming Language:** C++.
- ✓ **Code Editor:** Visual Studio Code.
- ✓ **Compiler:** C++ Compiler.
- ✓ **Storage:** Local files for storing records.

2.3 Hardware Requirement:

- ✓ Minimum 4GB RAM
- ✓ Dual-core processor.
- ✓ Compatible with desktops or laptops.

2.4 Training Requirement:

Basic understanding of:

- Computer operation.
- Menu-based applications.
- Different user roles and their functions.
- Basic C++ knowledge for maintaining the program.

2.5 Descriptions:

The School Management System consists of different modules:

- **Super Admin:** School and Principal management.
- **Principal:** Teacher and school management.
- **Teacher:** Student, attendance and result management.
- **Student:** Access to student-related information.
- **Parent:** Access to their child's information.

The system uses **C++ file handling** to store and retrieve different records.

2.6 Why C++?

- Supports Object-Oriented Programming.
- Provides efficient file handling.
- High performance and fast execution.
- Suitable for menu-driven applications.
- Supports classes, functions and other useful programming concepts.

CHAPTER 3: FEASIBILITY STUDY

3.1 Introduction:

A feasibility study is an evaluation done to determine whether a proposed project is practical and suitable for implementation.

3.2 Technical Feasibility:

Technical feasibility checks whether the required hardware, software and technology are available. The School Management System can be developed using C++ on a normal computer without requiring special hardware.

3.3 Economical Feasibility:

Economic feasibility checks whether the cost of developing the system is affordable. The project requires no expensive software or hardware and can be developed at very low cost.

3.4 Operational Feasibility:

Operational feasibility checks whether the system can be easily operated by its users. The School Management System provides a simple menu-based interface and requires only basic computer knowledge.

3.5 Conclusion:

The feasibility study confirms that the **School Management System is practical, affordable and suitable for implementation**.